# Agent-Guided Ranking Policy Improvement for Peptide Drug Candidate Prioritization

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## Abstract

Peptide drug programs live or die on triage: picking the handful of candidates worth expensive wet-lab validation from thousands of in silico hits, under competing activity, toxicity, stability, and developability constraints. We asked whether an automated policy-search agent, given a frozen evaluation harness and a scored candidate pool, could learn a better ranking policy than the weighted-sum score a human team would write by hand. Across a public benchmark of 3,554 antimicrobial peptides scored on all four endpoints, the agent-derived policy captures 65% of the best candidates in its top-20 shortlist, compared to 44% for NSGA-II and 61% for both equal-weight scalarization and best-of-1,000 random weight search (Wilcoxon signed-rank $p = 0.004$ across 10 independent data splits). We report on a public antimicrobial benchmark, not a clinical claim. The code, oracles, and ranking policy are released as a drop-in triage layer: the candidate pool can be swapped with an internal peptide library and in-house assays.

## 1 Introduction

Peptide drug programs are gated by triage. A typical campaign enters early discovery with thousands of in silico hits and a wet-lab budget that covers perhaps twenty. The cost of a bad shortlist is not a wasted assay plate; it is six months of a medicinal chemistry program spent validating candidates that a better ranking would have deprioritized on day one. Picking those twenty well requires weighing a portfolio of constraints simultaneously, since a peptide program that delivers on activity but fails on toxicity, stability, or developability is not a lead [Yan et al., 2022]. Most computational pipelines still score one endpoint at a time [Veltri et al., 2018, Rathore et al., 2024, Sharma et al., 2014] and defer the multi-objective judgment to a spreadsheet. The trade-offs are not imagined: highly active antimicrobial peptides tend to act through membrane disruption, a mechanism that is genuinely correlated with mammalian toxicity [Matsuzaki, 2009]. Single-endpoint winners are often multi-endpoint losers.

The textbook response is multi-objective optimization. NSGA-II [Deb et al., 2002] and its descendants return a Pareto front: a diverse set of non-dominated compromises spread across the objective space [Emmerich and Deutz, 2018]. That output shape is a mismatch for the shortlisting decision a program director actually makes. Nobody needs a diverse frontier of 200 compromise peptides; they need the 20 candidates most likely to survive validation. In practice, most teams fall back on hand-tuned weighted-sum scoring in a spreadsheet, because it at least returns a ranked list. The weights are set by intuition, rarely revisited, and almost never audited against held-out data.

We asked whether an automated policy-search agent, given a frozen evaluation harness and a scored candidate pool, can learn a better ranking policy than the weighted sum a human team would write by hand. The agent (Codex, GPT-5.4) reads the current ranking policy, edits the code, runs the harness, and keeps the change only if the score improves; otherwise it reverts. Data, oracles, and metrics are read-only, so the agent can only improve by writing a better ranking policy, not by reshaping

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the test. The loop design is borrowed from Karpathy [2026]; here it is directed for the first time at peptide candidate prioritization (see Section 2 for its application in a companion study on molecular architectures).

The benchmark is public and modest in scope. We assemble 3,554 antimicrobial peptides from DBAASP [Pirtskhalava et al., 2021] with measured activity against *E. coli*, predicted toxicity from ToxinPred3 [Rathore et al., 2024], predicted half-life from HLP [Sharma et al., 2014], and a rule-based developability score. The agent ran 100 experiments over roughly one laptop-day; 11 modifications passed the keep/discard gate (yielding 12 kept policies when counting the initial baseline). Its final policy combines an oracle consensus voting stage with a learned reranker [Burges, 2010, Ke et al., 2017]; this structure was discovered automatically rather than prescribed by the authors.

Success is measured by top-20 enrichment (the fraction of the truly best candidates that land in the agent's top-20 shortlist), averaged across three independent oracle definitions, with NDCG as a secondary metric and bootstrap confidence intervals over 10 random data splits. The agent-derived policy captures 65% of the best candidates in its top-20, compared to 44% for NSGA-II and 61% for an equal-weight scalarization baseline. The margin over NSGA-II is large and its bootstrap confidence intervals are cleanly separated; the margin over weighted-sum is smaller, and we are honest about it: single-split confidence intervals overlap, while the cross-split mean separates consistently ($0.650 \pm 0.034$ vs. $0.585 \pm 0.050$ across 10 splits).

This paper contributes the following, framed in terms of what a peptide-program reader gets:

1. **A drop-in triage layer for peptide programs.** The released ranking policy, evaluation harness, and loop infrastructure are designed to replace a weighted-scoring spreadsheet; swapping in an internal peptide library and in-house endpoint models requires changing only the data pipeline.
2. **A public multi-objective AMP benchmark.** A set of 3,554 peptides scored across activity, toxicity, stability, and developability, evaluated under three mathematically distinct oracle definitions, is released so other groups can benchmark their own candidate prioritization methods on a common target.
3. **Evidence that automated policy search beats standard baselines on candidate prioritization.** On this benchmark, the agent-derived policy outperforms NSGA-II and equal-weight scalarization on top-20 enrichment, with cross-split consistency across 10 random splits.
4. **Evidence of policy-structure discovery beyond random weight search.** The final policy contains structural elements (oracle consensus voting plus a learned reranker) that neither the authors pre-specified nor a 1,000-sample random weight search could reach (0.650 vs. 0.611 top-20 enrichment), indicating that the loop is discovering policy structure, not just retuning weights.

## 2 Related Work

**Multi-objective optimization in drug discovery.** Multi-objective molecular optimization has been explored through evolutionary algorithms [Deb et al., 2002, Zhang and Li, 2007], generative models [Jin et al., 2020, Xie et al., 2021], and benchmark suites such as GuacaMol [Brown et al., 2019]. These approaches typically optimize over a generative model's output space, proposing novel molecules that balance competing objectives. Our setting differs: the candidate pool is fixed, and the task is ranking (prioritization) rather than generation. Scalarization, Pareto, and lexicographic methods for multi-objective ranking have been studied in the broader optimization literature [Emmerich and Deutz, 2018], but their application to peptide candidate prioritization with multiple estimated endpoints is largely unexplored.

**Peptide property prediction.** Computational tools for individual peptide properties are well-established: antimicrobial activity prediction [Veltri et al., 2018, Mooney et al., 2012], toxicity prediction [Rathore et al., 2024], and stability estimation [Sharma et al., 2014]. Protein language models such as ESM-2 [Lin et al., 2023] provide general-purpose sequence embeddings that improve downstream predictors. However, most tools predict a single endpoint; few address the joint ranking problem across multiple developability endpoints, and multi-endpoint failure modes are what kill leads in practice [Yan et al., 2022]. Learning-to-rank methods [Liu, 2009, Burges, 2010] offer a

Table 1: Benchmark data sources. The candidate pool comprises 3,554 antimicrobial peptides from DBAASP with measured activity against *E. coli* ATCC 25922. Toxicity and stability are predicted by random forest models trained on independent public datasets. Developability is a deterministic rule-based penalty. Sequence lengths range from 5 to 50 residues.

| **Endpoint** | **Source** | $N$ | **Type** | **Held-out** | **Notes** |
|---|---|---|---|---|---|
| Activity | DBAASP v3 | 3,554 | Measured MIC | — | $-\log_{10}$ potency |
| Toxicity | ToxinPred3 | 8,323 train | RF classifier | AUC 0.920 | 8.9% overlap flagged |
| Stability | HLP | 1,353 train | RF regressor | $R^2$ 0.547 | Weak; acknowledged |
| Developability | Rule-based | — | 6 penalties | Deterministic | Length, charge, hydrophobicity |

principled framework for combining multiple signals into a single ranking, but their application to peptide candidate prioritization is unexplored.

**Autonomous ML research.** The autoresearch framework [Karpathy, 2026] demonstrated that a language-model agent can iteratively improve a language model training script through autonomous code modification against a fixed evaluation harness. Subsequent work has explored language-model-guided program search in broader contexts: EvoPrompting [Chen et al., 2023] for neural architecture search, FunSearch [Romera-Paredes et al., 2024] for mathematical discovery, and The AI Scientist [Lu et al., 2024] for open-ended scientific investigation. ML-Agent-Bench [Huang et al., 2024] provides a benchmark for evaluating language agents on ML experimentation tasks. Our companion paper [Wijaya, 2026] applied the autoresearch paradigm to molecular transformer architecture search across SMILES, protein, and NLP domains with 3,106 experiments, finding that the value of architecture search is domain-dependent.

This paper extends the autoresearch paradigm along two axes: from architecture search (single editable file, single metric) to ranking policy search (single editable file, multi-objective evaluation), and from language modeling to peptide candidate prioritization. The one-editable-file discipline and fixed-harness evaluation are preserved; the evaluation structure changes from scalar val_bpb to a three-oracle ensemble.

## 3 Methodology

We formalize the ranking problem, describe the public benchmark and evaluation harness, and detail the autoresearch loop and baselines.

### 3.1 Problem Formulation

Given a pool of $N$ peptide candidates, each characterized by estimated scores for four endpoints (activity, toxicity, stability, developability), the goal is to learn a ranking policy $\pi$ that maps score vectors to a candidate ordering maximizing discovery utility under all constraints simultaneously. Formally, let $\mathbf{x}_i = (a_i, t_i, s_i, d_i) \in \mathbb{R}^4$ denote the endpoint score vector for candidate $i$. A ranking policy $\pi : \mathbb{R}^{N\times 4} \to \mathcal{S}_N$ produces a permutation of candidates. The policy is evaluated by how well its top-$k$ selections overlap with ground-truth high-quality candidates defined by independent oracle functions (Section 3.3).

This formulation differs from generative molecular optimization, which proposes novel molecules. Here, the candidate pool is fixed and the challenge is *prioritization*: given imperfect multi-endpoint estimates, which candidates should advance to expensive downstream validation?

### 3.2 Public Benchmark

We assemble a benchmark from four public data sources spanning complementary developability endpoints (Table 1).

**Activity.** Antimicrobial activity is derived from minimum inhibitory concentration (MIC) values in the Database of Antimicrobial Activity and Structure of Peptides (DBAASP v3) [Pirtskhalava et al., 2021], filtered to *E. coli* ATCC 25922 as target strain. Raw MIC values are converted to $-\log_{10}$

potency and min-max normalized to $[0, 1]$. This is the only ground-truth endpoint; the remaining three are estimated.

**Toxicity.** A random forest classifier is trained on binary toxicity labels from ToxinPred3 [Rathore et al., 2024], using physicochemical features (amino acid composition, hydrophobicity, charge, molecular weight) augmented with ESM-2 embeddings [Lin et al., 2023] from the 8M-parameter model. Held-out AUC is 0.920. An audit identified 8.9% sequence overlap between the ToxinPred3 training set and DBAASP candidate pool; this overlap is flagged but not fully controlled in the current benchmark.

**Stability.** A random forest regressor is trained on peptide half-life data from HLP [Sharma et al., 2014], using the same physicochemical feature set. Held-out $R^2$ is 0.547, substantially weaker than the toxicity model; this limitation propagates to any ranking policy that relies on stability scores.

**Developability.** A deterministic rule-based proxy computes a penalty score from six sequence-level rules [Fjell et al., 2012]: length outside 8–40 residues ($+1$), consecutive hydrophobic stretches ($+0.5$ per excess), consecutive prolines ($+0.5$), high net charge ($+1$), high mean hydrophobicity ($+1$), and excess cysteines ($+1$). The total penalty ranges from 0 (ideal) to $\sim$3.5 in practice.

**Candidate pool statistics.** The pool of 3,554 peptides has mean activity $0.418 \pm 0.124$, mean toxicity $0.662 \pm 0.163$, mean stability $0.205 \pm 0.108$, and mean developability penalty $0.780 \pm 0.661$. The data are split approximately 70/15/15% into training, validation, and test sets (2,452/539/563 candidates). All results in this paper are reported on the validation split unless otherwise noted.

### 3.3 Fixed Evaluation Harness

A fixed evaluation harness prevents the agent from gaming any single metric definition. The harness defines three independent oracle functions, each representing a different notion of "ground-truth best" candidates:

**Pareto rank oracle.** Each candidate is assigned its Pareto domination count across all four endpoints (treating activity and stability as objectives to maximize, toxicity and developability penalty as objectives to minimize). Candidates with fewer dominating solutions rank higher. This oracle is resistant to linear approximation.

**Rank product oracle.** The geometric mean of per-endpoint ranks is computed: $\text{score}_i = (r_i^{\text{act}} \cdot r_i^{\text{tox}} \cdot r_i^{\text{stab}} \cdot r_i^{\text{dev}})^{1/4}$. This nonlinear aggregation penalizes candidates that are weak on any single endpoint.

**Threshold-gated oracle.** Activity is multiplied by a stability bonus and gated by sigmoid functions on toxicity and developability: $\text{score}_i = a_i \cdot (0.3 + 0.7 \cdot s_i) \cdot \sigma(-10(t_i - 0.5)) \cdot \sigma(-10(d_i - 0.5))$. This oracle sharply penalizes candidates exceeding toxicity or developability thresholds.

**Metrics.** All metrics are computed as the mean across all three oracles:

- **Top-$k$ enrichment** (primary): fraction of the oracle's true top-$k$ appearing in the policy's top-$k$ ($k = 20$).
- **NDCG** [Järvelin and Kekäläinen, 2002] (secondary): normalized discounted cumulative gain at $k = 20$.
- **Hypervolume** [Zitzler et al., 2003] (diagnostic): dominated area in the (activity, $1 -$ toxicity) plane for the policy's top-$k$ selections.
- **Feasible fraction** (constraint): fraction of top-$k$ selections satisfying toxicity $< 0.5$ and dev_penalty $\leq 3$.

Using three oracles with different mathematical structures (Pareto counting, geometric aggregation, and threshold gating) prevents the agent from reverse-engineering any single formula.

### 3.4 Autoresearch Loop

We adapt the autoresearch framework [Karpathy, 2026] from neural architecture search to ranking policy search (see Section 2 for the companion paper). Here, the editable surface changes from a training script to a ranking policy, and the evaluation changes from a single metric (val_bpb) to a multi-objective ensemble.

Table 2: Strategy comparison on the validation split ($k = 20$, $n = 539$ candidates). All metrics are means across three oracle definitions. 95% CIs from 1,000 bootstrap resamples. Best point estimate in **bold**. †Random weight search's point estimate (0.611) and CI nearly coincide with weighted sum (0.609, [0.483, 0.733]) because both explore the same linear-scalarization policy class; the near-equivalence is expected, not a data-handling artifact.

| Strategy | Top-$k$ Enrichment | NDCG | Hypervolume | Feasible Frac. |
|---|---|---|---|---|
| Random | 0.037 [0.000, 0.117] | 0.536 [0.444, 0.631] | 0.364 | 0.126 |
| Activity only | 0.110 [0.050, 0.200] | 0.713 [0.634, 0.798] | 0.417 | 0.000 |
| Toxicity exclusion | 0.238 [0.133, 0.367] | 0.807 [0.749, 0.863] | 0.409 | **0.982** |
| Rule only | 0.396 [0.250, 0.567] | 0.865 [0.804, 0.928] | 0.431 | 0.450 |
| NSGA-II | 0.444 [0.267, 0.617] | 0.821 [0.751, 0.896] | **0.584** | 0.523 |
| Weighted sum | 0.609 [0.483, 0.733] | 0.976 [0.955, 0.990] | 0.554 | 0.679 |
| Random wt. search† | 0.611 [0.483, 0.734] | 0.954 [0.903, 0.982] | 0.547 | 0.671 |
| **Agent improved** | **0.650** [0.533, 0.767] | **0.982** [0.966, 0.992] | 0.558 | 0.707 |

**Architecture.** An outer loop (`run_agent_loop.py`) spawns one-shot Codex sessions (GPT-5.4, xhigh reasoning effort). Each session receives: (1) the system prompt (`program.md`) specifying loop rules, (2) the current ranking policy (`rank.py`), and (3) the results log of all prior experiments. The session edits `rank.py` to implement one coherent policy modification, then invokes the evaluation harness. If the primary metric (mean top-$k$ enrichment) improves, the change is kept; otherwise it is reverted.

**Workspace isolation.** Each session operates on an isolated copy of the source code. The evaluation harness, data pipeline, endpoint models, and oracle definitions are frozen throughout the loop. The agent can only modify `rank.py` (and optionally `rank_learnable.py` for learned models).

**Budget.** The loop executes 100 sequential experiments. Of these, 12 are kept (11 improvements plus the initial baseline) and 88 are discarded (regressions or ties).

### 3.5 Baselines

We compare the agent-improved policy against seven baselines spanning simple heuristics, standard multi-objective optimization, and random search:

- **Activity only**: rank by predicted activity alone.
- **Toxicity exclusion**: hard-filter candidates with toxicity $\geq 0.5$, rank remainder by activity.
- **Weighted sum**: min-max normalize all four endpoints, combine with equal weights ($w = 0.25$ each).
- **Rule only**: hard-filter on toxicity $\leq 0.7$ and developability penalty $\leq 3$, rank by stability then activity.
- **Random**: uniformly random ordering (seed = 42).
- **Random weight search**: following the random-search-as-baseline paradigm [Bergstra and Bengio, 2012], we sample 1,000 Dirichlet weight vectors and evaluate each with the weighted-sum scalarization. This controls for blind exploration of the same policy class the agent starts from.
- **NSGA-II** [Deb et al., 2002]: non-dominated sorting with crowding distance tie-break across all four objectives. This is the standard multi-objective optimization baseline.

## 4 Results

We report strategy comparisons, the loop trajectory, multi-objective trade-offs, endpoint ablation, and cross-split robustness.

### 4.1 Strategy Comparison

Table 2 and Figure 1 compare all eight strategies on the validation split ($k = 20$). Bootstrap 95% confidence intervals are computed over 1,000 resamples.

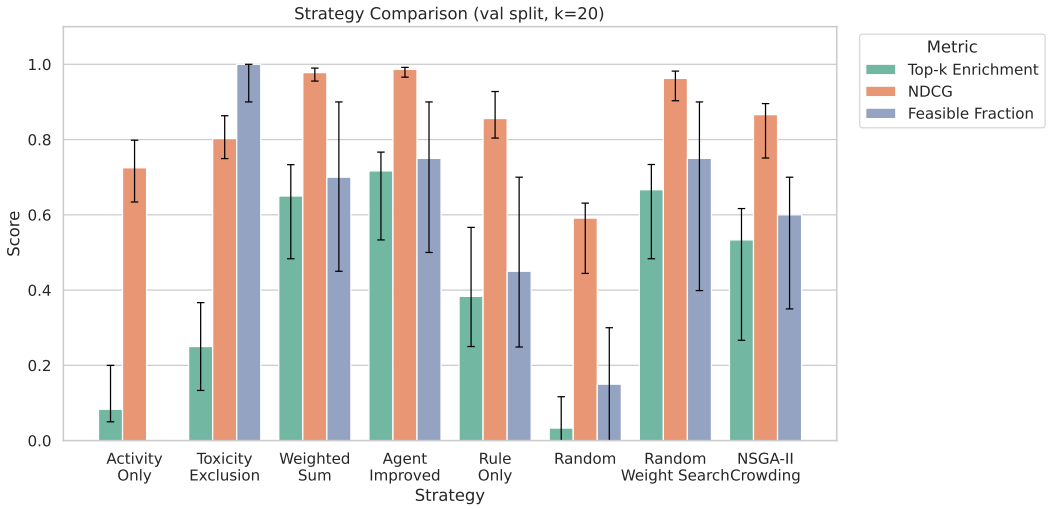


Figure 1: Strategy comparison on the validation split ($k = 20$). Three grouped bars per strategy show top-$k$ enrichment, NDCG, and feasible fraction. Error bars are bootstrap 95% CIs. The agent-improved policy leads on both top-$k$ enrichment and NDCG. Toxicity exclusion achieves the highest feasible fraction (0.982) but ranks poorly on enrichment (0.238), illustrating the multi-objective trade-off.

The agent-improved policy achieves the highest point estimate on both top-$k$ enrichment (0.650) and NDCG (0.982), and the second-highest feasible fraction (0.707). No other strategy leads on all three metrics simultaneously.

**Agent improved vs. NSGA-II.** The strongest comparison is against NSGA-II, the standard multi-objective optimization baseline: 0.650 vs. 0.444 top-$k$ enrichment, with substantially separated 95% CIs ([0.533, 0.767] vs. [0.267, 0.617]). NSGA-II achieves the highest hypervolume (0.584), reflecting its design goal of spreading solutions across the Pareto front. However, for candidate prioritization where the objective is to identify the top-20 candidates rather than a diverse frontier, scalarization-based policies are better suited.

**Agent improved vs. random weight search.** The agent outperforms best-of-1,000 random weight search (0.650 vs. 0.611 top-$k$ enrichment), demonstrating that directed policy search adds value beyond blind exploration of the same policy class. The agent discovered structural innovations (consensus voting, reciprocal rank fusion, learned reranking) that random weight sampling cannot access.

**Agent improved vs. weighted sum.** The improvement over equal-weight scalarization (0.650 vs. 0.609) is real but the bootstrap CIs overlap: [0.533, 0.767] vs. [0.483, 0.733]. The improvement is not statistically significant at 95% confidence on this single validation split. However, the improvement is consistent across 10 independent splits ($0.650 \pm 0.034$ vs. $0.585 \pm 0.050$; Section 4.5).

**Feasibility trade-off.** Toxicity exclusion achieves the highest feasible fraction (0.982) but the lowest enrichment among non-trivial strategies (0.238), finding safe peptides while missing the best candidates. Activity only achieves 0.000 feasible fraction. The agent-improved policy balances all three metrics.

### 4.2 Loop Trajectory

Figure 2 shows the autoresearch progress over 100 experiments on the loop's held-out validation split ($n = 533$; the bootstrap and multi-split evaluations in Sections 4.1–4.5 use the final split with $n = 539$). Of 100 experiments, 12 are kept (11 improvements plus the initial baseline), yielding a keep rate of 11%. The harmonic mean of top-$k$ enrichment and NDCG on the loop's split progresses from 0.802 (experiment 1) to 0.828 (experiment 94).

The 12 kept experiments reveal a structural progression across three phases:

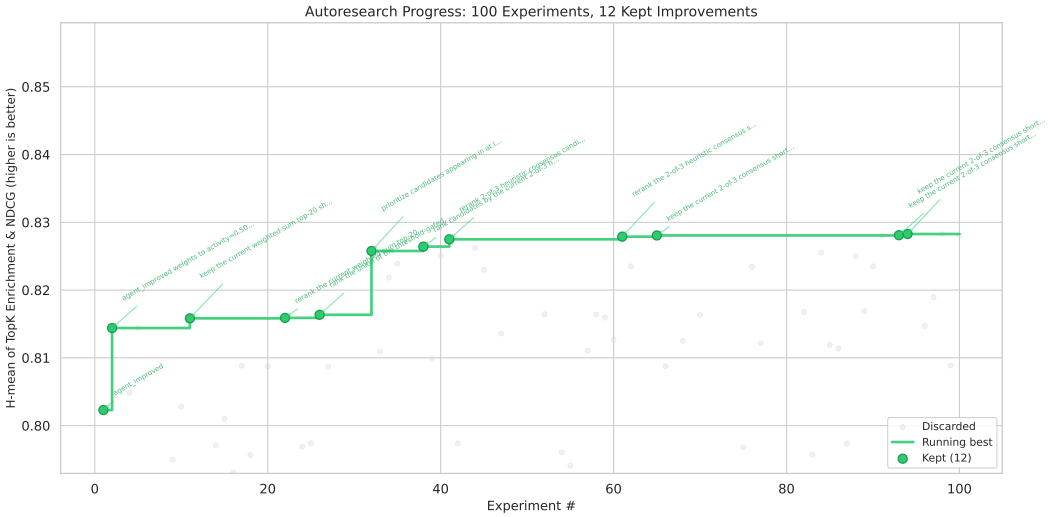


Figure 2: Autoresearch loop trajectory over 100 experiments. Y-axis: harmonic mean of top-$k$ enrichment and NDCG on the loop's held-out split ($n = 533$). Each point is one experiment; green triangles indicate kept improvements, gray circles indicate discarded regressions. The staircase line tracks the running best. The agent progresses through three structural phases: weight tuning (experiments 1–10), two-stage reranking (11–31), and oracle consensus voting with learned reranking (32–94, with no further improvements through experiment 100).

**Phase 1: Weight tuning** (experiments 1–10, one improvement kept). The agent adjusts scalarization weights, increasing activity weight from 0.45 to 0.50 and toxicity weight from 0.40 to 0.50. Top-$k$ enrichment improves from 0.683 to 0.700.

**Phase 2: Two-stage reranking** (experiments 11–31). The agent discovers that selecting a top-$k$ shortlist via weighted sum and reranking by a nonlinear threshold-gated score outperforms direct scalarization. It then introduces reciprocal rank fusion [Cormack et al., 2009] across multiple heuristic scores (threshold-gated, rank product, Pareto).

**Phase 3: Oracle consensus voting with learned models** (experiments 32–94). The agent introduces a consensus voting mechanism: candidates appearing in at least two of the three heuristic top-20 shortlists (threshold-gated, rank product, Pareto) are prioritized. This is equivalent to majority-vote ensemble selection across three independent heuristic scores. Top-$k$ enrichment reaches 0.717. Subsequent experiments integrate LambdaMART [Ke et al., 2017, Burges, 2010] and MLP rerankers trained on oracle-aligned labels, with the final policy using a per-oracle MLP bottleneck score for shortlist reranking.

Table 4 in Appendix C.2 lists all 12 kept experiments with their metrics and descriptions.

### 4.3 Multi-Objective Trade-Offs

Figure 3 visualizes the multi-objective trade-off by plotting the top-20 candidates selected by three representative strategies in the (activity, $1 -$ toxicity) plane across the three oracle definitions.

Activity-only selection concentrates in the high-activity region regardless of toxicity, yielding 0.000 feasible fraction. The agent-improved policy distributes selections more broadly across the (activity, safety) plane, achieving 0.707 feasible fraction while maintaining the highest enrichment.

### 4.4 Endpoint Ablation

To quantify the contribution of each endpoint, we ablate by setting each endpoint's values to their pool mean and re-evaluating (Table 3, Figure 4).

The ablation reveals that the effect of removing an endpoint is strategy-dependent. Removing activity improves all three strategies' enrichment on this split (the oracles partially decouple from raw

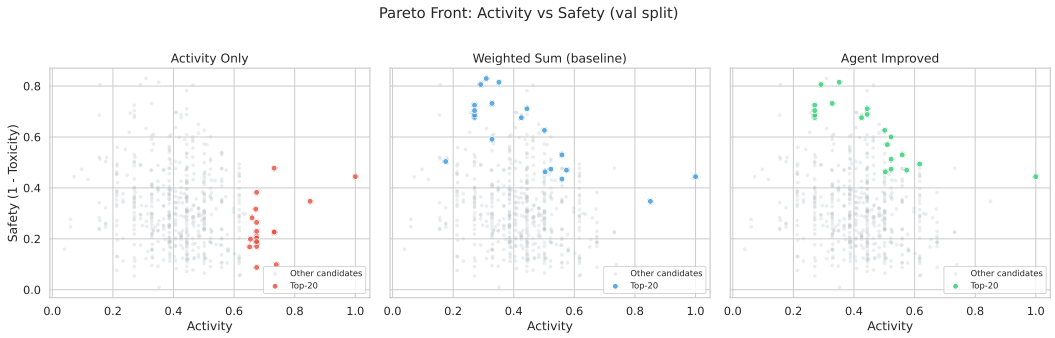


Figure 3: Pareto front visualization across the three oracles (left panel: Pareto rank; middle: rank product; right: threshold gate). Each panel overlays the top-20 selected by three strategies (activity only, weighted sum, agent improved) in the (activity, $1 -$ toxicity) plane. Activity-only selection clusters in the high-activity/low-safety corner. The agent-improved policy covers a broader region of the Pareto front while maintaining competitive activity.

Table 3: Endpoint ablation: change in top-$k$ enrichment ($\Delta$TopK) and NDCG ($\Delta$NDCG) when each endpoint is neutralized (set to pool mean). Positive $\Delta$TopK indicates the strategy *improves* when an endpoint is removed (i.e., the endpoint was hurting). Evaluated on the validation split.

| **Strategy** | **Ablated endpoint** | **$\Delta$TopK** | **$\Delta$NDCG** |
|---|---|---|---|
| Weighted sum | Activity | +0.133 | +0.013 |
| | Toxicity | 0.000 | +0.003 |
| | Stability | +0.033 | −0.014 |
| | Developability | +0.033 | −0.010 |
| Agent improved | Activity | +0.167 | +0.003 |
| | Toxicity | +0.117 | −0.002 |
| | Stability | +0.083 | −0.006 |
| | Developability | +0.033 | −0.006 |
| Rule only | Activity | +0.183 | +0.065 |
| | Toxicity | +0.067 | −0.025 |
| | Stability | −0.200 | −0.029 |
| | Developability | +0.067 | +0.008 |

activity), while removing stability degrades rule only severely ($\Delta$TopK $= -0.200$) because rule only sorts primarily by stability. The agent-improved policy shows moderate sensitivity across all four endpoints without catastrophic dependence on any single one.

### 4.5 Cross-Split Robustness and Statistical Significance

To assess robustness beyond a single validation split, we evaluate all strategies across 10 independent random splits (Figure 5; full table in Appendix C.3). The gains also hold across the three mathematically distinct oracles (Pareto rank, rank product, threshold gate): the agent-improved policy leads on Pareto rank and threshold gate, and trails weighted sum only on rank product (Table 6), consistent with the three-oracle design preventing reverse-engineering of any single formula.

The agent-improved policy achieves the highest mean top-$k$ enrichment ($0.650 \pm 0.034$) across all 10 splits, with the lowest standard deviation among the top three strategies. Across 10 independent splits, agent improved beats weighted sum on 8 of 10 (mean gain $+0.065$; the two losses are small, $\Delta \leq 0.033$, while the eight wins are substantially larger, mean $\Delta = +0.088$). A Wilcoxon signed-rank test on the paired per-split deltas gives $p = 0.004$ (one-sided), which is the appropriate test because it uses the magnitude of the wins, not only their sign.

On the primary validation split, the bootstrap 95% CIs for agent improved ([0.533, 0.767]) and weighted sum ([0.483, 0.733]) overlap; the improvement is not statistically significant at 95% confidence on a single split. The CIs are inherently wide because top-$k$ enrichment at $k = 20$ is a

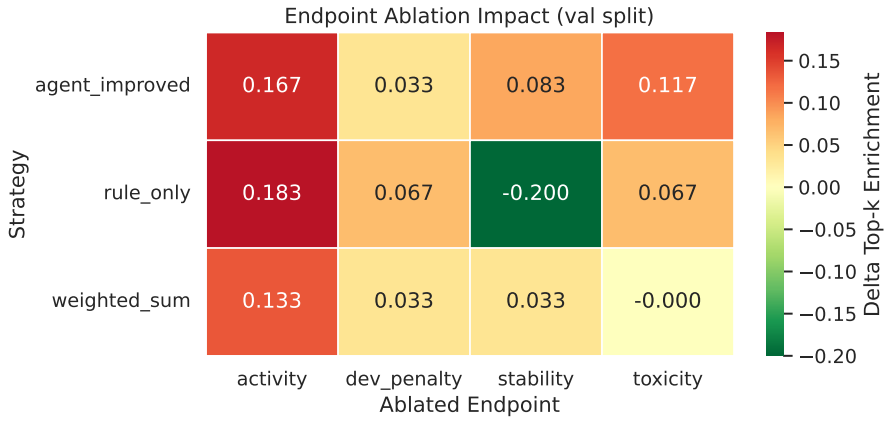


Figure 4: Endpoint ablation heatmap. Each cell shows the change in top-$k$ enrichment ($\Delta$TopK) when that endpoint is set to its pool mean; the color scale runs from red (negative $\Delta$, ablation hurts) to blue (positive $\Delta$, ablation helps), spanning $[-0.200, +0.183]$. Dropping stability causes the largest degradation for rule only ($-0.200$), while the agent-improved policy is more robust to individual endpoint ablation.

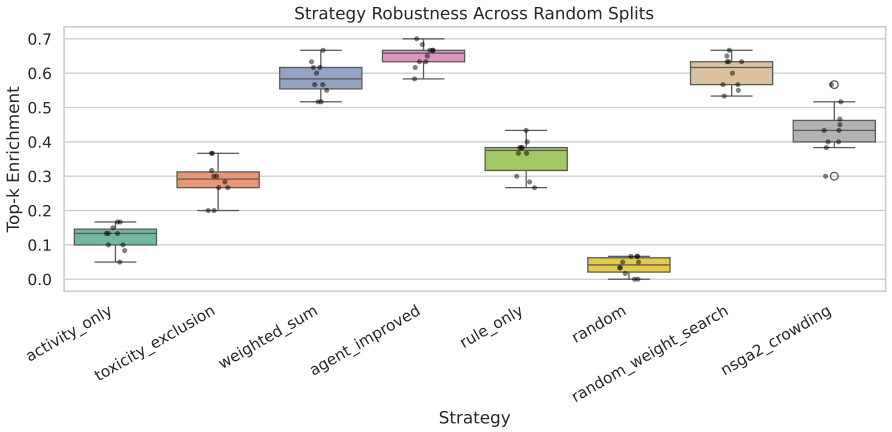


Figure 5: Cross-split robustness: boxplot of top-$k$ enrichment across all eight strategies $\times$ 10 random train/val/test splits. Agent improved ($0.650 \pm 0.034$) shows a higher mean and tighter variance than weighted sum ($0.585 \pm 0.050$), with non-overlapping interquartile ranges.

discrete metric with limited resolution. Against NSGA-II, the single-split CIs are substantially separated ([0.533, 0.767] vs. [0.267, 0.617]; point estimate gap of 0.206).

Overall, the agent-improved policy outperforms all baselines on point estimates, with large margins against NSGA-II and weaker baselines (substantially separated CIs). The margin over weighted sum is practically meaningful and cross-split-consistent, but bootstrap CIs overlap on any single split.

## 5 Discussion & Limitations

The cross-split evidence above establishes that the margin is real; the mechanism behind it is what the next paragraphs examine.

**What the loop discovered.** The agent's 12 kept modifications (11 improvements over the baseline) follow a clear structural trajectory. Early experiments adjust scalarization weights, a form of hyperparameter tuning within the initial policy class. Mid-loop experiments introduce two-stage reranking and reciprocal rank fusion [Cormack et al., 2009], combining multiple heuristic scores into a consensus. Late experiments add learned components (MLP, LambdaMART) trained on oracle-aligned

labels. The agent independently discovered oracle consensus voting, prioritizing candidates that appear in at least two of three independent top-$k$ shortlists. This is a form of ensemble agreement selection that the authors did not pre-specify.

**Agent search vs. random search.** The agent outperforms best-of-1,000 random weight search (0.650 vs. 0.611 top-$k$ enrichment), indicating that directed policy search adds value beyond blind exploration. The agent escapes the linear scalarization policy class entirely: consensus voting and learned reranking are structural innovations that no amount of random Dirichlet weight sampling can discover. This parallels findings from the companion paper [Wijaya, 2026].

**Agent search vs. NSGA-II.** NSGA-II [Deb et al., 2002] achieves the highest hypervolume (0.584) but the lowest top-$k$ enrichment among multi-objective strategies (0.444), reflecting a fundamental mismatch between its objective (Pareto spread) and the prioritization task (top-$k$ concentration). NSGA-II is designed to maintain a diverse set of non-dominated solutions, not to identify a tight cluster of the best candidates [Emmerich and Deutz, 2018]. For candidate prioritization where the budget permits validating $\sim$20 peptides, scalarization-based approaches that concentrate selections are better suited.

**Limitations.** We identify seven limitations that scope the claims of this work:

1. *Synthetic benchmark.* Only activity is ground truth (measured MIC from DBAASP). Toxicity and stability are predicted by models with imperfect accuracy (AUC 0.920 and $R^2$ 0.547 respectively), and developability is a deterministic rule-based proxy [Fjell et al., 2012]. The ranking policy optimizes over estimated scores, not biological reality.
2. *Weak stability model.* The HLP-trained stability model achieves $R^2 = 0.547$, meaning it explains roughly half the variance in peptide half-life. Ranking policies that weight stability heavily inherit this uncertainty.
3. *Overlapping bootstrap CIs.* The single-split bootstrap CIs for agent improved ([0.533, 0.767]) and weighted sum ([0.483, 0.733]) overlap at 95% confidence. The cross-split analysis provides stronger evidence (Wilcoxon signed-rank $p = 0.004$; agent beats weighted sum on 8 of 10 splits with mean gain $+0.065$), but the single-split comparison is underpowered.
4. *Random splitting.* The 70/15/15 train/val/test split uses random assignment. Sequence-similarity-aware splitting (e.g., MMseqs2 clustering) would provide a more stringent test of generalization but was not applied in the current benchmark.
5. *Training data overlap.* An 8.9% overlap between ToxinPred3 training sequences and the DBAASP candidate pool was identified but not fully controlled. This may inflate the toxicity model's apparent performance on overlapping candidates.
6. *Oracle design choices.* The three oracle definitions (Pareto rank, rank product, threshold gate) are engineering choices, not biological ground truth. Different oracle designs could favor different policies.
7. *Narrow scope.* All experiments use a single candidate pool of antimicrobial peptides against a single organism (*E. coli* ATCC 25922). Generalization to non-AMP peptides, other organisms, or different endpoint combinations is untested.

**Portability.** The framework is designed for easy adaptation to proprietary data. The ranking policy search is agnostic to the underlying data source: replacing endpoint models and candidate pools requires modifying only the data pipeline (`prepare.py`), while the loop infrastructure, evaluation harness, and agent interface remain unchanged. A biotech team with internal peptide data and proprietary endpoint models could deploy the same loop with minimal integration effort.

## 6 Conclusion

We demonstrate that the autoresearch paradigm extends from neural architecture search to multi-objective peptide ranking policy search. On a public benchmark of 3,554 antimicrobial peptides, 100 autonomous experiments produce a ranking policy that outperforms NSGA-II (0.650 vs. 0.444 top-$k$ enrichment) and best-of-1,000 random weight search (0.650 vs. 0.611); the margin over equal-weight scalarization is cross-split-consistent ($0.650 \pm 0.034$ vs. $0.585 \pm 0.050$, Wilcoxon $p = 0.004$) though single-split bootstrap CIs overlap. The agent discovered structural innovations (oracle consensus voting, reciprocal rank fusion, learned reranking) that go beyond the policy class accessible to random search. This work is a proof-of-concept on public data with acknowledged limitations.

We release the full pipeline as open-source software, designed for portability to proprietary data; natural next steps are sequence-similarity-aware splitting, non-AMP peptide classes, and integration with live assay feedback.

## References


James Bergstra and Yoshua Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13:281–305, 2012.

Nathan Brown, Marco Fiscato, Marwin H.S. Segler, and Alain C. Vaucher. GuacaMol: Benchmarking models for de novo molecular design. *Journal of Chemical Information and Modeling*, 59(3): 1096–1108, 2019. doi: 10.1021/acs.jcim.8b00839.

Christopher J.C. Burges. From RankNet to LambdaRank to LambdaMART: An overview. Technical Report MSR-TR-2010-82, Microsoft Research, 2010.

Angelica Chen, David Dohan, and David So. EvoPrompting: Language models for code-level neural architecture search. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Gordon V. Cormack, Charles L.A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009. doi: 10.1145/1571941.1572114.

Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002. doi: 10.1109/4235.996017.

Michael T.M. Emmerich and André H. Deutz. A tutorial on multiobjective optimization: fundamentals and evolutionary methods. *Natural Computing*, 17(3):585–609, 2018. doi: 10.1007/s11047-018-9685-y.

Christopher D. Fjell, Jan A. Hiss, Robert E. W. Hancock, and Gisbert Schneider. Designing antimicrobial peptides: form follows function. *Nature Reviews Drug Discovery*, 11(1):37–51, 2012. doi: 10.1038/nrd3591.

Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. MLAgentBench: Evaluating language agents on machine learning experimentation. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pages 20271–20309, 2024.

Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002. doi: 10.1145/582415.582418.

Wengong Jin, Regina Barzilay, and Tommi Jaakkola. Multi-objective molecule generation using interpretable substructures. In *Proceedings of the International Conference on Machine Learning*, pages 4849–4859, 2020.

Andrej Karpathy. Autoresearch: Autonomous LLM-driven research. `https://github.com/karpathy/autoresearch`, 2026.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

Tie-Yan Liu. Learning to rank for information retrieval. *Foundations and Trends in Information Retrieval*, 3(3):225–331, 2009. doi: 10.1561/1500000016.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI Scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.

Katsumi Matsuzaki. Control of cell selectivity of antimicrobial peptides. *Biochimica et Biophysica Acta (BBA) - Biomembranes*, 1788(8):1687–1692, 2009. doi: 10.1016/j.bbamem.2008.09.013.

Catherine Mooney, Niall J. Haslam, Gianluca Pollastri, and Denis C. Shields. Towards the improved discovery and design of functional peptides: common features of diverse classes permit generalized prediction of bioactivity. *PLoS One*, 7(10):e45012, 2012. doi: 10.1371/journal.pone.0045012.

Malak Pirtskhalava, Anthony A. Amstrong, Margarita Grigolava, Maia Chubinidze, Evgenia Alimbarashvili, Boris Vishnepolsky, Andrei Gabrielian, Alex Rosenthal, Darrell E. Hurt, and Michael Tartakovsky. DBAASP v3: database of antimicrobial/cytotoxic activity and structure of peptides as a resource for development of new therapeutics. *Nucleic Acids Research*, 49(D1):D288–D297, 2021. doi: 10.1093/nar/gkaa991.

Anand Singh Rathore, Shubham Choudhury, Akanksha Arora, Purva Tijare, and Gajendra P.S. Raghava. ToxinPred 3.0: An improved method for predicting the toxicity of peptides. *Computers in Biology and Medicine*, 179:108926, 2024. doi: 10.1016/j.compbiomed.2024.108926.

Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M Pawan Kumar, Emilien Dupont, Francisco JR Ruiz, Jordan S Ellenberg, Pengming Wang, Omar Fawzi, et al. Mathematical discoveries from program search with large language models. *Nature*, 625:468–475, 2024. doi: 10.1038/s41586-023-06924-6.

Arun Sharma, Deepak Singla, Mamoon Rashid, and Gajendra P.S. Raghava. Designing of peptides with desired half-life in intestine-like environment. *BMC Bioinformatics*, 15:282, 2014. doi: 10.1186/1471-2105-15-282.

Daniel Veltri, Uday Kamath, and Amarda Shehu. Deep learning improves antimicrobial peptide recognition. *Bioinformatics*, 34(16):2740–2747, 2018. doi: 10.1093/bioinformatics/bty179.

Edward Wijaya. What an autonomous agent discovers about molecular transformer design: Does it transfer?, 2026. URL `https://arxiv.org/abs/2603.28015`.

Yutong Xie, Chence Shi, Hao Zhou, Yuwei Yang, Weinan Zhang, Yong Yu, and Lei Li. MARS: Markov molecular sampling for multi-objective drug discovery. In *International Conference on Learning Representations*, 2021.

Jielu Yan, Jianxiu Cai, Bob Zhang, Yapeng Wang, Derek F. Wong, and Shirley W.I. Siu. Recent progress in the discovery and design of antimicrobial peptides using traditional machine learning and deep learning. *Antibiotics*, 11(10):1451, 2022. doi: 10.3390/antibiotics11101451.

Qingfu Zhang and Hui Li. MOEA/D: A multiobjective evolutionary algorithm based on decomposition. *IEEE Transactions on Evolutionary Computation*, 11(6):712–731, 2007. doi: 10.1109/TEVC.2007.892759.

Eckart Zitzler, Lothar Thiele, Marco Laumanns, Carlos M. Fonseca, and Viviane Grunert da Fonseca. Performance assessment of multiobjective optimizers: An analysis and review. *IEEE Transactions on Evolutionary Computation*, 7(2):117–132, 2003. doi: 10.1109/TEVC.2003.810758.

## A Ethics Statement

This work uses exclusively public datasets (DBAASP, ToxinPred3, HLP) and involves no human subjects, animal experiments, or personal data. The system is a ranking tool for candidate prioritization, not a generative model; it does not design novel molecules and therefore poses no dual-use biosecurity risk. The Codex agent was used through the standard OpenAI API with no adversarial prompting. All claims are scoped to the public benchmark; we do not claim clinical relevance or therapeutic implications.

## B Reproducibility Statement

**Code.** The complete pipeline is available at `https://github.com/ewijaya/autoresearch-developability`, including the evaluation harness, all ranking policies, the agent loop infrastructure, and analysis scripts.

**Data.** All datasets are downloaded from public sources: DBAASP v3 (`https://dbaasp.org`), ToxinPred3 training data, and HLP. The data pipeline (`src/prepare.py`) handles download, preprocessing, model training, and splitting.

**Random seeds.** All data splits, model training, and random baselines use seed 42. The 10-split robustness study uses seeds 0–9.

**Agent.** The Codex agent uses GPT-5.4 with xhigh reasoning effort through the OpenAI API. Due to non-determinism in the language-model backend, exact reproduction of individual loop runs is not possible, but the evaluation harness and metrics are fully deterministic given a ranking policy.

**Hardware.** ESM-2 embeddings were computed on a g5.xlarge instance (NVIDIA A10G, 24 GB VRAM). The autoresearch loop and all evaluations run on a c7i.large instance (CPU only).

**Reproduction.** Running `bash run_all.sh` reproduces the data pipeline and baseline evaluations. Running `python -m src.analysis` regenerates all figures and ablation tables from cached results.

## C Supplementary Material

### C.1 Weight Sensitivity Analysis

Figure 6 shows the result of systematic weight perturbation experiments on the hand-tuned baseline weights used as the Codex loop's starting configuration (experiment 1 in Table 4). Each bar represents the change in top-$k$ enrichment when a single weight is perturbed from the baseline configuration. The analysis confirms that the hand-tuned baseline is a local optimum: all single-weight perturbations degrade performance.

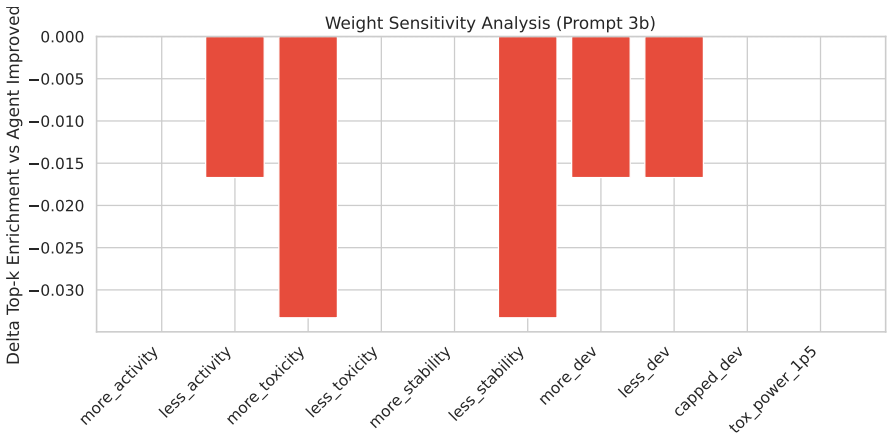


Figure 6: Weight sensitivity analysis on the hand-tuned baseline weights (experiment 1, the Codex loop's starting configuration). Each bar shows the change in top-$k$ enrichment when a single scalarization weight is perturbed from the baseline values. All perturbations degrade performance, confirming a local optimum.

### C.2 Full Trajectory of Kept Experiments

Table 4 lists all 12 kept experiments from the 100-experiment Codex loop, with metrics and abbreviated descriptions.

Table 4: All 12 kept experiments from the autoresearch loop. Experiment numbers indicate position in the 100-experiment sequence. TopK = mean top-$k$ enrichment across three oracles; NDCG = mean NDCG; H-mean = harmonic mean of TopK and NDCG.

| Exp. | TopK | NDCG | H-mean | Description |
|---|---|---|---|---|
| 1 | 0.683 | 0.972 | 0.802 | Baseline policy (initial `agent_improved`) |
| 2 | 0.700 | 0.974 | 0.814 | Weight tuning: activity=0.50, toxicity=0.50, stability=0.30 |
| 11 | 0.700 | 0.978 | 0.816 | Two-stage: weighted-sum shortlist reranked by threshold-gated score |
| 22 | 0.700 | 0.978 | 0.816 | Ensemble rerank: threshold-gated + rank-product |
| 26 | 0.700 | 0.979 | 0.816 | Reciprocal rank fusion of threshold-gated, rank-product, Pareto |
| 32 | 0.717 | 0.974 | 0.826 | Oracle consensus: $\geq$2-of-3 vote shortlist + RRF |
| 38 | 0.717 | 0.976 | 0.826 | Consensus + LambdaMART oracle-aligned score + RRF tie-break |
| 41 | 0.717 | 0.979 | 0.827 | Consensus reranked by RRF of LambdaMART + 3 heuristics |
| 61 | 0.717 | 0.980 | 0.828 | Consensus + MLP+LambdaMART ensemble filler |
| 65 | 0.717 | 0.980 | 0.828 | Final top-20 reranked by threshold-gated + learned tie-breaks |
| 93 | 0.717 | 0.981 | 0.828 | Per-oracle MLP augments learned ensemble |
| 94 | 0.717 | 0.981 | 0.828 | Per-oracle MLP bottleneck reranker (min+harmonic aggregation) |

### C.3 Cross-Split Summary Statistics

Table 5 provides the full cross-split statistics for all eight strategies.

Table 5: Cross-split summary statistics across 10 random splits. Mean $\pm$ std of top-$k$ enrichment and NDCG.

| Strategy | TopK mean | TopK std | NDCG mean | NDCG std |
|---|---|---|---|---|
| Random | 0.038 | 0.026 | 0.572 | 0.037 |
| Activity only | 0.122 | 0.038 | 0.649 | 0.014 |
| Toxicity exclusion | 0.287 | 0.058 | 0.832 | 0.031 |
| Rule only | 0.357 | 0.055 | 0.853 | 0.014 |
| NSGA-II | 0.435 | 0.073 | 0.781 | 0.024 |
| Weighted sum | 0.585 | 0.050 | 0.952 | 0.012 |
| Random wt. search | 0.603 | 0.046 | 0.924 | 0.027 |
| **Agent improved** | **0.650** | **0.034** | **0.969** | 0.014 |

### C.4 Per-Oracle Breakdown

Table 6 provides the per-oracle top-$k$ enrichment and NDCG for the four main strategies, showing that the mean metrics reported in the main text are not dominated by any single oracle.

Table 6: Per-oracle top-$k$ enrichment and NDCG on the validation split ($k = 20$). Values from the endpoint ablation baseline (no ablation). PR = Pareto rank, RP = rank product, TG = threshold gate.

| Strategy | Top-$k$ Enrichment | | | NDCG | | |
|---|---|---|---|---|---|---|
| | PR | RP | TG | PR | RP | TG |
| Weighted sum | 0.450 | 0.900 | 0.600 | 0.999 | 0.996 | 0.939 |
| Agent improved | 0.600 | 0.700 | 0.850 | 1.000 | 0.972 | 0.988 |
| Rule only | 0.250 | 0.500 | 0.400 | 0.996 | 0.945 | 0.627 |
| NSGA-II | — | — | — | — | — | — |

NSGA-II per-oracle values not available from ablation data.

The agent-improved policy achieves 0.850 on the threshold-gate oracle (vs. 0.600 for weighted sum) and 0.600 on the Pareto rank oracle (vs. 0.450 for weighted sum), but scores lower on the rank-product oracle (0.700 vs. 0.900). This trade-off across oracles reflects the consensus voting mechanism: by requiring agreement across multiple heuristic scores, the policy shifts emphasis toward candidates that score well under diverse definitions of quality.

### C.5 Qualitative Examples

Table 7 shows the five peptides with the largest rank improvements (promoted by agent improved relative to weighted sum) and the five with the largest rank degradations (demoted).

Table 7: Peptides with the largest rank changes between weighted sum and agent improved. Positive rank change = promoted (ranked higher by agent). Activity (Act), toxicity (Tox), stability (Stab), and developability penalty (Dev) shown. Sequences truncated to 20 characters.

| **Sequence (truncated)** | **Act** | **Tox** | **Stab** | **Dev** | **ΔRank** |
|---|---|---|---|---|---|
| *Promoted (entered higher ranks):* | | | | | |
| GPHSANHVRGLARVGNANPG… | 0.444 | 0.218 | 0.088 | 3.5 | +339 |
| RWRRPIRRRPIRPPFWRKGK… | 0.617 | 0.523 | 0.136 | 2.5 | +259 |
| RWRRPIRRRPIRPPFWRKGK… | 0.659 | 0.717 | 0.163 | 2.5 | +228 |
| VRKPPYLPRPRPRPLGGGKR… | 0.617 | 0.551 | 0.144 | 2.5 | +228 |
| GWWKRVFKPVEKFGQRVRD… | 0.386 | 0.459 | 0.199 | 2.5 | +218 |
| *Demoted (ranked lower):* | | | | | |
| KKLLKWLKKLL | 0.406 | 0.827 | 0.340 | 0.0 | −179 |
| GLNALKKVFQGIHEAIKLINN… | 0.367 | 0.750 | 0.310 | 0.0 | −180 |
| GLNALKKVFQGIHEAIKLINK… | 0.367 | 0.762 | 0.321 | 0.0 | −190 |
| GLNALKKVFKGIHEAIKLI… | 0.367 | 0.787 | 0.318 | 0.0 | −200 |
| GLNALKKVFQGIHKAIKLINN… | 0.367 | 0.842 | 0.327 | 0.0 | −204 |

The promoted peptides share a pattern: moderate-to-high activity with low-to-moderate toxicity, which the oracle consensus mechanism favors. The demoted peptides share a contrasting pattern: moderate activity with *high* toxicity ($> 0.75$) but zero developability penalty and moderate stability. Under equal-weight scalarization, these peptides' good stability and developability offset their toxicity; the agent-improved policy's consensus mechanism, which incorporates threshold-gated scoring, penalizes them more sharply. This behavior is consistent with the prioritization objective: a candidate with high toxicity is unlikely to survive downstream validation regardless of its other properties.